

# Mohamed Ashfaq

Computer Science Undergraduate | Full-Stack, AI & IoT Developer  
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## PROFILE

**Full-Stack & AI-Integrated Developer** and Computer Science undergraduate with a strong track record of independently designing, building, and deploying end-to-end web applications, AI-powered systems, and ML models using **Java (Spring Boot), React, PostgreSQL, LLM APIs, and TensorFlow Lite**. Experienced in **REST API design, JWT authentication, CI/CD pipelines**, and Agile/Scrum workflows — with **8 shipped projects** spanning full-stack web, IoT, edge AI, and machine learning. Strong theoretical foundation in **deep learning (LSTM, CNN, ANN), ensemble methods, performance metrics, KNN, Naive Bayes, Decision Trees, and regression** — backed by hands-on coursework and end-to-end ML pipelines. Committed to writing clean, maintainable code and delivering production-ready software..

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, TypeScript, C++, HTML5, CSS3

**Back-End / Databases:** Spring Boot, MySQL, PostgreSQL, REST API, JWT Authentication, Microservices

**Front-End Frameworks:** React, Next.js, Flutter

**DevOps & Tools:** Docker, CI/CD (GitHub Actions), Git, GitHub, Version Control, Postman, Figma, Electron

**Testing:** Unit Testing, JUnit, H2 In-Memory Database, Automated Testing

**Data & ML:** Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Jupyter Notebooks

**Core Concepts:** OOP, DSA, Agile/Scrum, Distributed Systems, IoT, Machine Learning, Team Leadership, Problem Solving

## PROJECTS & EXPERIENCE

### Airea — Smart Respiratory Monitor for Early Lung Cancer

Sep 2025 – Apr 2026

GitHub: <https://github.com/Siddiha/Airea> | Live: <https://airea.lk>

C++ (ESP32), Flutter, Spring Boot, Supabase, FCM, TensorFlow | *Freelance Team Project*

- Engineered an **end-to-end IoT health monitoring platform** over **7 months** with a **6-member team** — covering ESP32 wearables, a **Spring Boot REST API** backend, and a Flutter app with Supabase storage and FCM real-time alerts.
- Deployed TensorFlow Lite Micro models** on ESP32-S3 for on-device cough detection (CNN, INT8 quantized) and fall detection — eliminating cloud round-trips and enabling real-time edge inference with sub-second latency.
- Designed the full **system architecture** — bridging embedded C++ firmware on ESP32, a **Spring Boot microservice** backend, and a cross-platform Flutter mobile app into a single cohesive platform.

### Ai - MediBook — Clinic Appointment System

Mar 2026 – Apr 2026

GitHub: <https://github.com/ashfaq-ui/MediBook> | Live: <https://medi-book-flax.vercel.app/>

React, Spring Boot, PostgreSQL, JWT, GitHub Actions, Vercel, Railway | *Freelance Project*

- Built **solo in under 6 weeks** — a full-stack clinic system with **3 user roles** (Patient, Doctor, Admin), **15+ REST API endpoints**, JWT authentication, BCrypt hashing, and Gmail SMTP notifications.
- Set up **CI/CD pipeline** via GitHub Actions with **automated unit testing** (H2 in-memory); deployed to Vercel and Railway — reducing manual deployment effort to zero.
- Built a **React** frontend with **protected routes**, role-specific dashboards, and responsive UI for all 3 user roles — connecting to the **Spring Boot REST API** with Axios.
- Integrated Google Gemini API** to power a multi-turn Patient Health Assistant and one-click Doctor Appointment Summary — both served via a custom Spring Boot AI layer.

### Invoice Manager — Desktop Finance & Sales Ledger App

Mar 2026 – Apr 2026

GitHub: <https://github.com/ashfaq-ui/invoice-manager>

Electron, JavaScript, Node.js | *Freelance Project*

- Delivered **solo in 3 weeks** — cross-platform desktop app for Mac and Windows managing invoices, sales ledger, and financial analysis across **3 report types** with zero-install distribution via GitHub Releases.

## Loan Approval Prediction — Machine Learning

Jan 2026 – Apr 2026

GitHub: <https://github.com/ashfaq-ui/Machine-learning-cw>

Python, Scikit-learn, Pandas, NumPy, Matplotlib, Plotly, Jupyter | Academic Project

- Built an end-to-end **ML pipeline** on **58,640 samples**; KNN tuned via GridSearchCV achieved **92.21% accuracy** and **0.9519 AUC-ROC** — the best performing model across 3 classifiers.
- Performed full **data preprocessing** — outlier removal, missing value imputation, StandardScaler feature scaling — and implemented **Voting Ensemble** and pruned **Decision Tree Regressor** with feature importance analysis.

## SpendWise — Distributed Full-Stack Expense Tracker

Feb 2026 – Mar 2026

GitHub: [https://github.com/ashfaq-ui/spend-wise\\_](https://github.com/ashfaq-ui/spend-wise_) | Live: <https://jazzy-concha-e4ad62.netlify.app/>

Java (Spring Boot), React.js, PostgreSQL, REST API | Freelance Project

- Developed **solo in 4 weeks** — a 3-tier personal finance app with **10+ REST API endpoints**, PostgreSQL persistence, and a React frontend with real-time updates; deployed and live on Netlify.
- Designed a normalized **PostgreSQL schema** for transaction and category management; implemented full **CRUD operations** with **Spring Boot JPA** repositories.

## Black Olives — Premium Apple Retail E-Commerce Platform

Jun 2025 – Sep 2025

GitHub: <https://github.com/ashfaq-ui/webpro1-nextjs> | Live: <https://web-pro-1.netlify.app/>

Next.js | Freelance Project

- Built **solo in 4 months** — a full **Next.js SSR** e-commerce platform with product catalogue, dynamic routing, and SEO-optimized pages; live and deployed.
- Leveraged **Next.js getServerSideProps** and dynamic routes to pre-render product pages — improving **page load performance** and search engine indexability.

## EDUCATION

### BSc (Hons) in Computer Science

Sep 2024 – Present

Informatics Institute of Technology (Affiliated with University of Westminster), Colombo

- Level 04 Average: 76.8% | Notable: Computer System Fundamentals, Mathematics for Computing, Software Development II — Java
- Level 05 (In Progress) Current Average: 80% : Database Systems, OOP, SDGP, Machine Learning & Data Mining, Algorithms, Client Server Architecture

### Primary & Secondary Education

Jan 2010 – Dec 2023

Hindu College, Rathmalana

- G.C.E. Advanced Level — Physical Science: Combined Mathematics & Physics.
- G.C.E. Ordinary Level: Strong results including distinction in Mathematics.

## ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Hackathon Finalist — CodeRally 6.0 and Xtreme 19.0, focusing on rapid development and problem-solving under pressure.
- Competitor — Hult Prize competition and SLIIT MATHFEST 2023.
- IEEE Member at Informatics Institute of Technology (2024 – Present).
- Leadership — Prefect (2019–2022) and Event Coordinator at Hindu College, Rathmalana.

## CERTIFICATIONS

- Learning Docker (LinkedIn Learning, 2026)
- TypeScript for JavaScript Developers (LinkedIn Learning, 2025)
- React Essential Training (LinkedIn Learning, 2025)
- Critical Thinking and Problem Solving (LinkedIn Learning, 2024)
- Python GUI Development with Tkinter | Learning Python (LinkedIn Learning, 2024)

## REFERENCES

### Aakash Nambiyarooran

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